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This is a report of a regression analysis on the mtcars dataset. This report focuses on examining how fuel efficiency relates to key automobile characteristics. We perform a correlation analysis and point estimate as well as an interval estimate of the coefficients of three simple linear regression models. We present our findings below.

1. The relationship between mpg, as a response variable and **drat**, **disp**, **hp**, and **wt** turn out to be interesting due to the fact that there seems to be a linear relationship in the scatterplots, that is to say that the scatter plots suggest that we could use any of these four variables to predict the mpg variable. This suggestion is consistent with the values of the correlation coefficients between the mpg variable and each of these covariates.

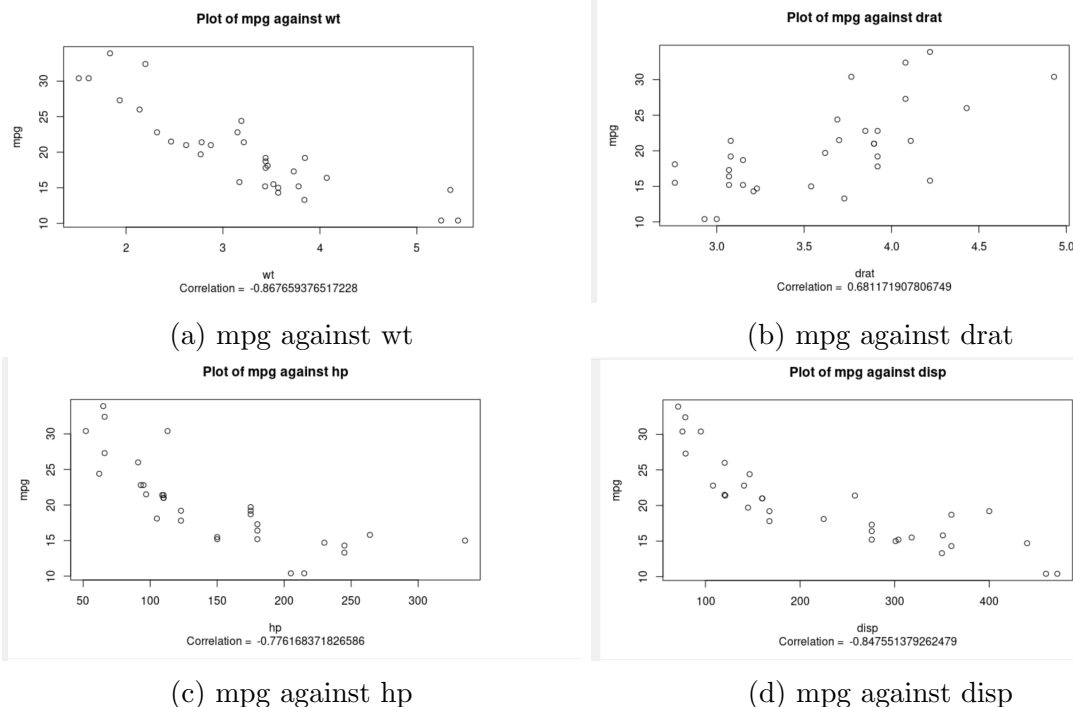


Figure 1: Plots of interesting relationships

2. We notice that for each of the three linear regression models we reject the null hypothesis for each estimate except the estimate for the intercept of the model where the drat variable

is a covariate. Notice that this conclusion is consistent with the 95%—confidence intervals calculated for each estimate in each model. which are given below:

model	95%—confidence interval for β_1	95%—confidence interval for β_0
disp model	$(-0.05083797, -0.03159227)$	$(27.08843246, 32.11127705)$
drat model	$(4.60113, 10.755335)$	$(-18.70946, 3.660219)$
wt model	$(-6.486308, -4.202635)$	$(33.450500, 41.119753)$

Table 1: Table of confidence intervals

Thus all but the estimate for the intercept in the drat model are significant. For the other significant estimates, we see the estimate of the slope is negative in the disp model thus a unit increase in the disp variable causes a decrease of about 0.04 units in the mpg variable. A similar situation is observed in the case of the wt variable. The negative slope indicates that a unit increase in the values of the wt variable cause a decrease of about 5.3 units of the mpg variable. The estimate of the slope of the drat variable is positive indicating that a unit increase in drat variable results in an increase of about 7.6 units in mpg values. For the significant intercept estimates, we have that for the disp model, when the disp variable is at 0 then the mpg variable is estimated to have a value of about 29.5 units and for the wt variable, we estimate that a 0 value of the wt variable results in a value of 37.2 units of the mpg variable.

3. For the model with disp as the covariate, we see that the residuals form a rough cloud of data around the zero line, which is the assumed mean, with roughly the same amount of points above the line as below which is good since we assumed independent error terms. We do not notice any pattern of the residuals.

For the model with drat as covariate, we see a similar trend like before where the data points form a cloud of data around the zero line, with the same number of points above the line as below. We also do not notice any pattern in the residual plot.

For the model with wt as covariate, we see a similar trend where the number of data points below the zero line is roughly the same as those above it.



(a) residuals of mpg vs disp (b) residuals of mpg vs drat (c) residuals of mpg vs wt

Figure 2: Plots of Residuals

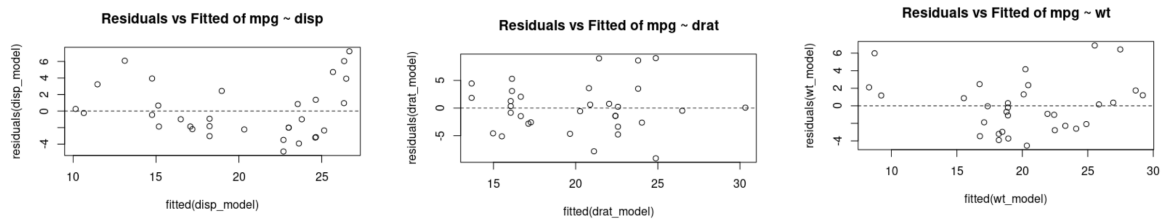
4. For $\text{disp} = 290$ we get a predicted value of mpg to be 17.64747. This input value falls within the range of data observed for the disp variable.
For $\text{drat} = 3.25$ we get a predicted value of mpg to be 17.42964. This input value falls within the range of data observed for the drat variable.
For $\text{wt} = 2.950$ we get a predicted value of mpg to be 21.51894. This input value falls within the range of data observed for the wt variable.

5. Based on the formulas derived in class, we get the estimated values of β_0, β_1 and σ^2 for the model with *disp* as covariate as: $\hat{\beta}_0 = 29.59985, \hat{\beta}_1 = -0.04121512, \hat{\sigma}^2 = 10.57196$, with *drat* as covariate as $\hat{\beta}_0 = -7.524618, \hat{\beta}_1 = 7.678233, \hat{\sigma}^2 = 20.11889$, and with *wt* as covariate respectively $\hat{\beta}_0 = 37.28513, \hat{\beta}_1 = -5.344472, \hat{\sigma}^2 = 9.277398$. In all three cases these values are consistent with the values estimated when we fit the model with R's inbuilt method.

6. For the case where *drat* is the covariate we have: the 95% confidence interval for the intercept is $(-18.70946, 3.660219)$ and the 95% confidence interval for the slope is $(4.60113, 10.755335)$. For the both the slope and intercept, based on the available data, these intervals provide plausible ranges for the true parameter values based on the data. However for the intercept, since the range includes 0 then it is possible the mpg has a value of 0 when the drat value is 0, in addition, the larger range indicates some uncertainty in our estimate. But for the slope, we know that the drat variable affects the mpg variable since the specified bounds for the confidence interval of the slope do not contain 0 and it affects it positively since the interval specified contains only positive values.
 For the case where *disp* is the covariate we have: the 95% confidence interval for the intercept is $(27.08843246, 32.11127705)$ and the confidence interval for the slope is $(-0.05083797, -0.03159227)$. For the both the slope and intercept, based on the available data, these intervals provide plausible ranges for the true parameter values based on the data. For the intercept, since the range does not include 0 then the intercept is significant and likewise for the slope, since the range does not include 0, then the slope is significant and negative.
 For the case where *wt* is the covariate we have: the 95% confidence interval for the intercept is $(33.450500, 41.119753)$ and the 95% confidence interval for the slope is $(-6.486308, -4.202635)$. For the both the slope and intercept, based on the available data, these intervals provide plausible ranges for the true parameter values based on the data. Over here too, in both cases we see that neither interval includes 0 thus both slope and intercept are significant.

7. In all three cases the linear model is appropriate. This is because in particular we notice the negative linear trend in both the *disp* and *wt* variables and a positive linear trend in the *drat* variable. This is also supported by the correlation coefficient in each case. Additionally, the plots of the residuals against the fitted values have roughly the same number of points below the mean line as above the mean line that is at residual equal to 0, except in the case of the *disp* model where the number of points above the mean line are less than the number of points below the line, although not by much. There is no pattern in each of the plots for each model, we just observe a cloud of data points. which is consistent with the model assumption of uncorrelated error terms. In addition, we notice a roughly symmetric interval, about the 0 value, of dispersion of the residual points. In particular in the case of the *disp* covariate, the residuals are spread roughly around $(-6, 7)$ and in the case of the *drat* covariate the models are spread roughly around $(-7, 6)$ and in the case of the *wt* covariate the residuals are spread around $(-6, 7)$. These roughly symmetric intervals of dispersion indicate a roughly constant variance, which is consistent with our assumptions that the error terms in the true relation have a constant variance. We observe a few outliers in the residual plot of the *drat* model against the fitted mpg values at residual values greater than 5. This means that our fitted model performed poorly for those particular data points hence it may be necessary to run the

analysis without those data points to observe what happens.



(a) Residuals vs Fitted of mpg vs disp (b) Residuals vs Fitted of mpg vs drat (c) Residuals vs Fitted of mpg vs wt

Figure 3: Plots of Fitted vs Residuals